

Question number	Answer	Additional guidance	Mark
7(a)(i)	A description that includes four of the following points: • {identify/isolate/ cut out} the (<u>human</u>) rHE gene (1) • Cut {DNA sequence / gene} using restriction {endonuclease / enzyme} (1) • insert the gene into a vector (1) • inserting vector into suitable target (<u>sheep</u>) {cell / tissue} (1)	gene coding for rHE is isolated using restriction enzymes = 2 marks accept named vector e.g. plasmid, virus ignore bacteria e.g. udder cells / milk (protein) producing cells / fertilized sheep egg cell / embryo / zygote	Expert (4)

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7(a)(ii)	77.5 (%)	accept 77 or 78	Graduate (1)

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7(a)(iii)	A description that includes four of the following points: - rHE acts as an antigen (1) {(phagocyte / macrophage / APC) presents antigen to (B/T) lymphocyte (1) - T helper cells activate the B cells (1) - differentiate into plasma cells (1) - plasma cells {release/ produce} (anti-rHE) antibodies (1)	accept antigen recognized as foreign accept B cells present antigen to themselves. accept T helper cells cause the production / formation of B effector cells and B memory cells	Expert (4)

Question number	Answer Additional guidance	Mark
7(b)	An answer that includes the following points: Indicative content - description from graphs and experimental data • (many) crops have been engineered to provide resistance to pests or herbicides eg. soybean, corn • soybean will not be killed by (glyphosphate) herbicides • corn will not be eaten by the core borer • ----- • herbicide tolerant soybean went from 7% of planted area in 1996 to 95% in 2017 • insect resistant corn went from 6% in 1996 to 88% in 2017 • increase in planting of both GM soybean and GM corn • more GM soybean planted (than GM corn) • general increase in production of genetically modified crops ----- explanation of benefits • higher yields / less damage by pests • more appealing colour / useful feature • GE foods having longer shelf life • more predictable food supplies for ever growing population • ----- • improved nutritional content • ‘medication’ provided through GE foods • can help prevent {malnutrition / disease} • ----- • (reduced use / cost) of herbicides and pesticides • withstand environmental conditions • ----- discussion of risks • increasing demand agricultural resources and land • reducing land available to grow other crops • ----- • increasing monoculture • declining biodiversity / stated effect on (stated local ecosystem)	Expert (6)

resistant

To go in published MS

Level 1:

1 mark - table or graph commented on

2 marks - both graph and table commented on

Level 2:

all level 1 plus:

basic discussion of both benefits and risks (only 1 section)

OR detailed discussion of EITHER benefits OR risks

Level 3:

all level 2 plus:

detailed discussion of BOTH benefits AND risks

selective and concise and logical

